Q11: Why Do LLMs Hallucinate?

DLS 2026 — Module (Paradise)

October 2026

Learning objectives

After this module, a student should be able to (i) define hallucination and distinguish it from lying and from honest reasoning errors, (ii) state the compression/generalization, calibration, and knowledge-boundary accounts of why fluent falsehood emerges from next-token prediction, (iii) explain the Kalai–Vempala result that calibrated models must hallucinate on arbitrary facts, (iv) evaluate standard mitigations (retrieval, post-training) together with their documented failure modes, and (v) name the main evaluation instruments and what each actually measures.

1 What hallucination is, and is not

A language model *hallucinates* when it produces output that is fluent and confident but unfaithful to its source, its training data, or the world. The standard survey distinguishes *intrinsic* hallucination (output contradicting the source) from *extrinsic* hallucination (output unverifiable from the source), and catalogues causes from noisy training data to erroneous parametric knowledge and exposure bias [2]. Three clarifications keep the term honest:

1. **Hallucination is not lying.** Lying requires intent to deceive — a model of the reader's beliefs and a goal of exploiting it. A next-token predictor has neither; it samples from a learned distribution. The failure is epistemic, not moral. This matters practically: you cannot deter a statistical process by demanding honesty, only by changing the process.
2. **Fluency is not evidence.** The same machinery that produces grammatical, coherent text produces its falsehoods. Surface quality carries no information about truth value, so humans who treat fluency as a credibility signal are systematically misled — precisely what TruthfulQA measures in humans' own misconceptions replicated by models [7].
3. **Not every error is a hallucination.** A multi-step arithmetic slip where each step is a plausible local continuation is a reasoning failure, not a fabricated fact. Conflating them produces bad post-mortems: the fixes differ.

A working taxonomy for this module:

- **Fabrication:** the output has no source — a cited paper that does not exist, an API function that was never written.
- **Misattribution:** every component is real — a real author, a real venue, a real year — but the combination is false. The most common form in citations, biographies, and structured answers.
- **Faithful error:** the model's premises were right and its reasoning slipped — wrong count, wrong unit, dropped negation.

The distinction is operational, not merely stylistic: fabrication tracks unattested content, misattribution tracks interference between attested contents, and faithful errors track inference, so each responds to different interventions.

2 Account 1: Compression and generalization

Pre-training is lossy compression of the corpus under a parameter budget. A model that memorized every fact would generalize poorly; a model that generalizes must abstract, and abstraction means the specific truth of a rare fact is traded away for the statistics of common cases. The model's objective rewards *plausible continuation*, and truth is only one subclass of plausible: the corpus itself contains fiction, error, contradiction, and out-of-date claims, so a perfect predictor of the corpus would still emit falsehoods. TruthfulQA formalizes this as *imitative falsehood*: questions where the corpus's popular answer is false ("What happens if you crack your knuckles a lot?"). Early models reproduced human misconceptions at a high rate — the best tested model was truthful on 58% of questions against a 94% human baseline, with *larger* models generally *less* truthful on this benchmark [7].

The same pressure has a quantitative face: QA accuracy correlates with how often a fact's evidence appears in pre-training data, and long-tail facts are not rescued by scale — models would need to grow by many orders of magnitude to answer rarely-attested questions reliably [4]. Hallucination is thus partly a *frequency phenomenon*: the learned distribution concentrates on the popular continuation, and the true continuation of a rare fact loses the argmax to a more plausible rival. The toy experiment in `code/hallucination_toy.py` makes this concrete: a smoothed count model trained on a 100% truthful corpus still answers rare facts with the most popular entity, and its accuracy climbs a clean monotone staircase as a function of training frequency (0.00 at counts 0–2, 0.44 at counts 3–5, 1.00 at counts 6+).

3 Account 2: Calibration — hallucination as sampling beyond the reliability boundary

An LLM is a distribution over continuations; generation is a *sampling* act, not a lookup. Even a perfectly *calibrated* model — one whose token probabilities match true conditional frequencies — samples from that distribution, and any sample that is not the single most-supported claim risks being false. Call the set of contexts where the model's support for the truth exceeds its support for every rival the *reliability boundary*: greedy decoding is correct only inside it, and sampling is correct only where the truth's probability mass dominates, a strictly smaller region. Hallucination is what falling outside that boundary looks like from outside.

Kalai and Vempala make this precise [3]. Idealize facts as *arbitrary* — their truth cannot be inferred from training data alone ("Alex Wilkins ate at Salumeria on a Tuesday") — versus *systematic* (arithmetic, or facts whose truth is checkable by rule). For arbitrary facts, any calibrated model with bounded per-fact probability must hallucinate at a rate lower-bounded by its *monofact rate*: the fraction of training facts that appear exactly once, the Good–Turing estimator of the missing-fact mass. The intuition: there are exponentially more plausible-but-false arbitrary factoids than true ones; a calibrated model must spread probability over unattested candidates, and that spread *is* the hallucination rate. Three hedges matter. (i) The bound applies to arbitrary facts only — it does not force hallucination on systematic facts or frequently-attested references, which the authors attribute to capacity and learning dynamics instead. (ii) It assumes ideal training data; real corpora are noisier. (iii) It is a statement about *pre-training* to be a good predictor — post-training

can trade calibration for reduced hallucination, which is exactly what alignment pipelines observe empirically. The theorem demystifies: some hallucination is not a bug of Transformers but a price of calibrated prediction under an open world.

4 Account 3: Knowledge boundaries — parametric memory vs. context

A third lens: the model carries two knowledge sources with different authorities — *parametric memory* (facts baked into weights during pre-training) and *context* (the prompt, retrieved documents, conversation state). Hallucination frequently arises at their boundary: the context implies one answer, the parameters insist on another, and the resolved output satisfies neither. When context is absent, the parametric answer for a rare fact is simply the popular continuation — the frequency effect above. Interpretability work locates parametric associations in mid-layer MLP modules, editable in place by rank-one edits (ROME), which is evidence that factual recall is a localized, look-up-like computation that can fail independently of fluency [9].

Post-training interacts with this boundary in a documented way: fine-tuning on facts the model does not already know teaches it that *generating unsupported claims is acceptable behavior*. Gekhman et al. show such "new-knowledge" examples are learned far more slowly than knowledge-consistent ones, and that as they are eventually memorized they raise hallucination on held-out questions roughly linearly [1]. The supported conclusion is modest but important: fine-tuning best teaches the *form* of answers, not new world knowledge; new facts should enter through the data (pre-training or retrieval), not through SFT.

5 Mitigations and their honest limits

Retrieval-augmented generation. RAG conditions generation on documents retrieved from an external index, combining parametric and non-parametric memory [5], and retrieval-in-the-loop dialogue measurably reduces hallucination over parametric-only baselines [10]. But retrieval moves the problem, it does not delete it: the retriever can fetch irrelevant or wrong passages, the generator can ignore correct ones, and Mallen et al. show retrieval *hurts* on popular entities the model already knows — the retrieved context misleads a model whose parametric answer was right [8]. Grounding to a bad source produces confident, citable falsehood. The failure mode of a RAG system is inherited from its index, and no amount of generation-side fluency audits it.

Post-training. Alignment reduces (does not eliminate) hallucination, at a measured cost: pipelines optimized for factuality degrade calibration — the very property whose loss Kalai–Vempala's bound flags [3]. RLHF-tuned models are systematically overconfident; a refusal or hedge, being low-probability under the base distribution, must be installed against the gradient of the training distribution.

Decoding and verification. Constrained decoding, self-consistency, and sample-and-verify filter the distribution after the fact; they shift the operating point on the reliability boundary but cannot move the boundary for unattested facts, and verifiers can themselves hallucinate.

No mitigation is a cure; each converts one failure mode into another with a different signature. Engineering reality is choosing which failure signature you can audit.

6 Evaluation

- **TruthfulQA** [7]: 817 adversarial questions targeting imitative falsehoods; measures resistance to *popular* wrong answers. Limitations: saturation risk under contamination, static, multiple-choice and generation variants.
- **Factual-association probes** [9]: cloze templates over (subject, relation, object) triples with causal-tracing edits; measure *parametric storage*, not end-to-end truthfulness. They answer "does the model contain the fact," not "will it say the fact."
- **HaluEval** [6]: 35K crowd- and model-generated hallucinated/non-hallucinated pairs across QA, dialogue, summarization; measures *recognition* of hallucination in context, where strong models are far from ceiling. Limitations: partially synthetic labels, English-centric.

No single number captures "hallucination rate"; each instrument measures one slice (imitative falsehood, parametric storage, recognition), and progress on one does not certify the others. A system owner should track the slice that matches their failure mode — fabrication-heavy workloads need retrieval audits, not TruthfulQA scores.

7 Synthesis

The three accounts are one story told at three levels. *Compression*: truth is a subset of plausibility, and rare truths lose to frequent plausibilities. *Calibration*: a calibrated predictor must spread mass over the unattested, so sampling emits falsehoods at a rate the monofact count predicts. *Knowledge boundaries*: the popular continuation wins precisely where parametric memory is thin, and context, when wrong or ignored, fails to override it. Mitigations are boundary management — retrieval extends what counts as attested; post-training trades calibration for caution; verification filters samples. The module's code demonstrates the first two accounts in miniature, with the same corpus driving every number: the model is perfectly truthful *in distribution* (100% of its argmax answers are attested, real entities) yet systematically false on rare facts, and its confidence crosses 50% only for the most-attested facts — the reliability boundary of a toy Kalai–Vempala world.

What to remember. Fluency is free; truth is not. The same loss that makes an LLM usable makes it fallible, and the fallibility is predictable from corpus statistics — which is the optimistic part: a failure you can predict from frequency data is a failure you can route around with retrieval, abstention, and audit.

References

[1] Zorik Gekhman, Gal Yona, Roee Aharoni, Matan Eyal, Amir Feder, Roi Reichart, and Jonathan Herzig. Does fine-tuning llms on new knowledge encourage hallucinations? In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 7765–7784, 2024.

[2] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12), 2023.

[3] Adam Tauman Kalai and Santosh S. Vempala. Calibrated language models must hallucinate. In *Proceedings of the 56th Annual ACM Symposium on Theory of Computing (STOC)*, pages 160–171, 2024.

[4] Nikhil Kandpal, Haikang Deng, Adam Roberts, Eric Wallace, and Colin Raffel. Large language models struggle to learn long-tail knowledge. In *Proceedings of the 40th International Conference on Machine Learning*, 2023.

[5] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

[6] Junyi Li, Xiaoxue Cheng, Wayne Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. Halueval: A large-scale hallucination evaluation benchmark for large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6449–6464, 2023.

[7] Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3214–3252, 2022.

[8] Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Hannaneh Hajishirzi, and Daniel Khashabi. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9802–9822, 2023.

[9] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in gpt. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

[10] Kurt Shuster, Spencer Poff, Moya Chen, Douwe Kiela, and Jason Weston. Retrieval augmentation reduces hallucination in conversation. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 3784–3803, 2021.